Section -4

ISOLATOR METALLICS

1.0 SCOPE: Refer Clause-1 of Section-I, General

2.0 STANDARDS:

Unless otherwise specified elsewhere in this specification. The rating as well as performance and testing of the Isolators shall confirm to the latest revisions and amendments of the following standards available at the time of placement order.

Sl. No.	Standard	Title
1.	IS: 9921	Specification for Alternating current Isolators (Isolators) and Earthing switches for voltages above 1000 V
2.	IEC: 129	Alternating current Isolators and Earthing switches
3.	IS:2147	Degree of protection provided by enclosure for low voltage switch gear and control gear.
4.	IS:4691	Degree of protection provided by enclosure for rotating electrical machinery
5	IS:12063	Classification of degrees of protection provided by enclosures of electrical equipment
6.	IS:2629	Recommended practice for hot dip galvanising of iron and steel.
7.	IS:4759	Hot dip Zinc coatings on structural steel and other allied products.
8.	IS:2633	Method for testing uniformity of coating of Zinc coated articles.
9.	IS:6735	Fastners – Spring Lock washers for screws with cylindrical heads.
10.	IS:2016	Plain washers.
11.	IEC-71	Electrical clearances
11.	IEC-71	Electrical clearances
12	IS-1573	Electroplated coating of Zinc on iron and steel

REFERENCE STANDARDS	NAME AND ADDRESS
IS	Indian Standard Bureau of India Standards Manak Bhavan 9, Bahadurshah Zafar Marg, New Delhi - 110 002, INDIA
IEC	International Electro Technical Commission Bureau Central de la Commission, Electro Technique International 1 Rue de verembe, Grneva,Switzerland
ISO	International Organisations for Standardization Danish Board of Standardization Danish Standardizing Sraat Aurehoegvaj - 12,DK-2900 Heelprup DENMARK

Equipment meeting with the requirements of other authoritative National or International Standards, which ensure equal or better performance than the standards mentioned above, shall also be considered. When the equipment offered by the tenderer conforms to other standards salient points of difference between standards adopted and the standards specified

in this specification shall be clearly brought out in the relevant schedule. Four copies of such standards with authentic translation in English shall be furnished along with the offer.

3.0 CLIMATIC CONDITIONS: Refer Clause-2 of Section-I, General

4.0 PRINCIPAL PARAMETERS :

The equipment covered in this specification shall meet the technical requirements listed below:

Sl. No.	Technical Parameter	220kV 2500/2000/ 1600/1200/ 800A Isolator	132 kV 800A/1600A Isolator	33kV 2000A Isolator	33 kV 800A Isolator
1.	Rated Frequency (Hz)	50	50	50	50
2.	System Neutral Earthing	Effectively Earthed			
3.	No. of Poles	3	3	3	3
4.	Temperature Rise	As per Relevant IS/IEC Publication			
5.	Safe duration of overload a) 150% of rated current(minutes) b) 120% of rated current(minutes)	 5 30	 5 30	 5 30	 5 30
6.	Rated voltage (kV rms) / Highest system voltage(kV rms)	220/245	132/145	33/36	33/36
7.	Type of Isolator	Double Break			
8.	Rated normal current(A) rms	800/2500/ 2000/1600/ 1200/	800/1600A	1200/ 2000A	800
9.	Rated short time withstand current (kA) rms of Main Switch & Earth Switch for 3 Sec. duration	50	40	31.5	31.5
10.	Rated dynamic withstand curent (kA)	125	100	78.75	78.75
11.	Basic insulation level: i) 1.2/50 micro second lightning impulse withstand voltage(+ve or -ve polarity)				
	a. To Earth (kVp)	±1050	±650	±170	±170
	b. across isolating distance One terminal subjected to lightning impulse(kVp) and opposite terminal subjected to power frequency (kV) rms Voltage (as per IS)	±1200	±750	±195	±195
	ii) Rated 1 minute power frequency withstand voltage(kV) rms				
	a Across isolating distance(kV) rms	530	315	80	80

	b. To earth and between poles (kV) rms	460	275	70	70
12.	Min. creepage distance of support and rotating insulator (mm)	7595 (31mm)	4495 (31mm)	1116 (31mm)	1116 (31mm)
13.	Phase to phase spacing for installation (mm)	4500	3000	1300	1300
14.	Minimum clearances (mm) :				
	a. Safety Clearance	2500	2500	2500	2500
	b. Live part to ground	4900	3800	2820	2820
15.	Rating of auxiliary contacts	10 A at 220 V DC with breaking capacity of 2 A DC with time constant not less than 20 ms.			
16.	Operating type	12 Sec./less	Manual	Manual	Manual
17.	Rated mechanical terminal load	As per relevant standards			
18.	Rated magnetizing/ capacitive current make/ break (A) rms	0.7	0.7	0.7	0.7
19.	RIV at 1 MHz & 1.1 x rated phase to earth voltage	1000 micro volts	--	-	-
20.	Operating mechanism of				
	Isolator	A.C. Motor operated & Manual operated	A.C. Motor operated & Manual operated	A.C. Motor operated & Manual operated	
	Earth switch	Manual and Motor operated	Manual and Motor operated	Manual /Motor operated	Manual / Motor operated
	Type of operation	Gang operated	Gang operated	Gang operated	
	Type of control	Local /Remote	Local /Remote	Local /Remote	Local /Remote
21.	No of Auxiliary contacts				
	On each isolator	12 NO + 12 NC	12 NO + 12 NC	4 NO + 4 NC	4 NO + 4 NC
	On each earthing Switch	6 NO + 6 NC	6 NO + 6 NC	4 NO + 4 NC	4 NO + 4 NC
22.	a.Rating of auxiliary contacts	10A, 220V DC	10A, 220V DC	10A, 220V DC	10A, 220V DC
	b.Breaking Capacity of auxiliary contacts	2A DC with circuit time constant of not less than 20 ms	2A DC with circuit time constant of not less than 20 ms	2A DC with circuit time constant of not less than 20 ms	2A DC with circuit time constant of not less than 20 ms

23.	Operating time	12 sec. or less	12 sec. or less	NA for manual operation	
24.	No. of terminals in control cabinet	All contacts and control circuits are to be wired up to control cabinet and additional 24 Nos. spare terminals exclusively for purchaser's use All contacts and control circuits are to be wired up to control cabinet and additional 24 Nos. spare terminals exclusively for purchaser's use			
25.	Local remote switch indication of Isolator for Substation Automation system (SAS)	Shall be provided	Shall be provided	Shall be provided(as per site requirment)	Shall be provided(as per site requirment)
26.	Minimum Cantilever strength (upright) in kg	1000kg	600kg		
27.	Mechanical Endurance class	Isolator-M2	E/S-M0	Isolator-M1	
27.	Type of supporting Insulator	Solid core post type	Solid core post type	Solid core post type	Solid core post type
	a) Top flange - Pitch Circle Diameter	127 +/-0.5	127 +/-0.5	76+/-0.5	76 +/-0.5
	b) Bottom flange - Pitch Circle Diameter	225+/-1.0	184 +/- 1.0	76+/-0.5	76 +/-0.5
	c) Height of insulator stack mm	2300 +/-3.5	1500 +/-2.5	508 +/-1.0	508 +/-1.0
	d) Height of each basic unit mm	2300 +/-3.5	1500 +/-2.5	508 +/-1.0	508 +/-1.0

5.0 TECHNICAL REQUIREMENTS :

5.1 The 220kV, 132 kV and 33kV double break Isolators shall be a set of three individual poles with one vertical break-earthing blade per pole suitable for fixing on either side of the poles or without any earthing blade (applicable for 132 kV Isolators).The 33 kV Isolators shall be overhead type only.

5.2 Purchaser reserves the right to choose all Isolators of any one type specified or any combination of the same with earth switch or without earth switch. Bids would however, be analyzed based on quantities of each type as specified in Section-VI of this specification.

5.3 Complete Isolator with all the necessary items for successful operation shall be supplied including but not limited to the following:

5.4 Isolators assembled with isolator blades (main and earth), complete base frame, linkages, Operating mechanism complete etc.,

5.5 In case of 220kV and 132 kV class Isolators central control cabinet for each isolator with all the required electrical devices mounted therein and the necessary terminal blocks for termination shall be provided.

5.6 Material of earthing blades and contacts shall be same as those of main blades and contacts. Cross sectional area of earthing blades and contacts shall not be less than 50% of cross sectional area of main blades and contacts. The earthing blades shall have the same short time current rating (thermal and dynamic) as that of main blades.

5.7 OPERATING MECHANISM:

5.7.1 a) Each 220kV Isolator shall be remote controlled from the control room. Provision shall be made for local motor control also. Operating mechanism shall also be equipped with local manual operating device intended for emergency operation when motor operating mechanism becomes inoperative. Additional electromagnetic type interlock shall be provided so as to prevent the operation of Isolator manually or electrically when the corresponding circuit breakers is On.

b) The operating mechanism of 132 kV Isolators shall be equipped with manual operating device, through crank and reduction gear Provision shall be made for motor control also and for 33 kV Isolators the operating mechanism shall be with manual operating device and provision shall be made for motor control also. It shall be possible to pad lock the manual operating handle both in the open and close positions of the switches. Additional electromagnetic type interlock shall be provided so as to prevent the operation of Isolators when the corresponding circuit breaker is 'ON'. Earth switches also shall be manually operated by separate operating mechanism.

c) Manual operating mechanism through crank and reduction gear shall be provided for isolator earth switch.

5.7.2 The operating mechanism shall provide quick simple and effective operation. The operating mechanism shall be suitable for operation of all three poles simultaneously i.e. gang operation . The design of manual operation shall be such that one man shall be able to operate the isolator without undue effort with about 20 (Twenty) revolutions of the crank. The operating mechanism shall be suitable to hold the Isolator in CLOSE OR OPEN position and prevent operation by gravity, wind, short circuit forces, seismic forces, vibration, shock, accidental touching etc., limit switches for control shall be fitted on the Isolator/ earth switch. Shaft within the cabinet to sense the open or close positions of the isolators and earth switches.

5.7.3 It shall not be possible, after final adjustment has been made, for any part of the mechanism to be displaced at any point in the travel sufficient enough to cause improper functioning of the isolator when the same is opened or closed at any speed. All holes in cranks, linkage etc., having moving pins, shall be drilled to fit accurately so as to maintain the minimum amount of slack and lost-motion in the entire mechanism.

5.7.4 A ‘local/remote selector switch and a set of open/close push buttons shall be provided on the control cabinet of the isolator to permit its operation through local or remote push buttons. The remote push buttons shall be arranged by the purchaser.

5.7.5 Provision shall be made in the control cabinet to disconnect power supply to prevent local/remote power operation.

5.7.6 Each motor operated mechanism shall be subjected to blocked rotor test.

5.8 CONTROL CABINET :

The control cabinet of theoperating mechanism shall be made out of 12 SWG (2.64 mm thick) sheet steel or 3 mm thick aluminium plate or casting. Hinged door shall be provided with pad locking arrangement. Sloping rain hood shall be provided to cover all sides. 15 mm thick neoprene or better type of gaskets shall be provided to ensure degree of protection of at least IP55 as per IS:2147. the cabinet shall be

suitable for mounting on support structure with adjustment for vertical alignment. Details of these arrangements shall be furnished along with the tender. The drum pipe may be provided to suit the structure height of 2860mm for 220kV.

5.8.1 The motor shall be suitable

5.9 GEAR:

The Isolator may be required to operate occasionally, with considerably long idle intervals. special care shall be taken for selection of material for gear and lubrication of gears to meet this requirement. The gears shall be made out of aluminium bronze or any other better material and lubricated for life with graphite or better quality non-draining and non hardening type grease. Wherever necessary automatic relieving mechanism shall be provided. Complete details of components, material, grade, self lubricating arrangement, grade of lubricants, details of jig, fixtures and devices used for quality check, shall be furnished by the bidder in his offer.

5.10 GLAND PLATE AND GLANDS:

A removable gland plate with double compression type brass cable glands shall be provided with each operating mechanism for terminating all cables. Exact quantity of glands to be provided, shall be intimated to the supplier.

5.11 AUXILIARY SWITCH :

5.11.1 Each operating mechanism box shall be equipped with Twelve Nos. NO + Twelve Nos. NC for 220kV, Twelve Nos. of NC and Twelve nos. of NO for 132kV and four Nos. of NC and four nos. of NO for 33kV, auxiliary switches exclusively for Purchaser's use and NO/NC contacts shall be provided in the auxiliary switch box for earth mechanism as per principle parameters.

5.11.2 The auxiliary switches and auxiliary circuits shall be capable of carrying the current of at least 10 Amps continuously.

5.11.3 Auxiliary switches shall be capable of breaking at least 2 A in a 220 V D.C. circuit with a time constant of not less than 20 milliseconds.

5.11.4 Quick make and break (QMB) type auxiliary switch shall have snap action built-in within the switch.

5.11.5 The auxiliary switches shall be actuated by a cam or similar arrangement directly mounted on the isolator and shall be without any intermediate levels, linkages etc., to ensure fool proof operation.

5.12 TERMINAL BLOCK AND WIRINGS :

Each operating mechanism shall be provided with 1100 V grade stud type terminal block. All auxiliary switches, interlocks and other terminals shall be wired upto terminal block. The terminal block shall have at least 20% extra terminals. All wiring shall be carried out with 1100 V grade insulated 2.5 sq. mm. copper wires.

5.13 DUTY REQUIREMENTS :

5.13.1 Isolators and earth switches shall be capable of withstanding the dynamic and thermic effects of the maximum possible short circuit current of the systems in their closed position. They shall be constructed such that they do not open under influence of short circuit current.

5.13.2 The earth switches, wherever provided, shall be constructionally interlocked so that the earth switch can be operated only when the main switch is open and not vice-

versa. The constructional interlocks shall be built-in construction of isolator and shall be in addition to the electrical and mechanical interlocks provided in the operating mechanism.

5.13.3 In addition to the constructional interlock, isolator and earth switches shall have provision to prevent their electrical and manual operation unless the associated and other interlocking conditions are met. All these interlocks shall be of fail safe type. Suitable individual interlocking coil arrangements shall be provided. The inter locking coil shall be suitable for continuous operation from DC supply and within a variation range stipulated in this volume.

5.13.4 The isolator shall be capable of making/breaking normal currents when no voltage occurs across the terminals of each pole of isolator on account of make/break operation.

5.13.5 The isolator shall be capable of making/breaking magnetising current of 0.7 A at +0.15 power factor and capacitive current of 0.7 A at -0.15 power factor.

5.13.6 Isolator and earth switches shall be able to bear on the terminals the total forces including wind loading and electrodynamic forces on the attached conductor without impairing reliability or current carrying capacity.

5.14 INTERIOR ILLUMINATION :

A switch, HRC fuse and a holder suitable for a 240 V lamp shall be provided in the control cabinet.

5.15 ACCESSORIES :

The accessories to be provided on the isolator shall include but not limited to the following :

5.16 POSITION INDICATOR :

A position indicator to show whether the isolator is in ON or OFF position

5.16.1 COUNTER BALANCE SPRINGS :

Counter balance springs, cushions etc. shall be provided to prevent impact at the end of travel both on opening and closing of the isolator. The springs shall be made of durable and non-rusting type alloy.

5.17 NAME PLATES :

Isolator and earthing switches and their operating devices shall be provided with a name plate. The name plate shall be weather proof and corrosion proof. It shall be mounted in such a position that it shall be visible in the position of normal service and installation. It shall carry the information as per the relevant standards along with name of APTRANSCO as Transmission Corporation of A.P.Limited duly engraved or punched on it.

5.18 Padlocking Device:

All Isolator and earthing switches shall be provided with padlocking device to permit locking of the isolator and earthing switch in both fully open and fully closed positions.

5.19 Signaling:

5.19.1 Signaling of the close position shall not take place unless the movable contact has set in a position in which the rated normal current, the peak withstand current and the short time withstand current can be carried safely.

5.19.2 Signaling of the close position shall not take place unless the movable contact has reached the position such that the clearance between the movable and fixed contacts is atleast 80% of the isolating distance.

5.20 Earthing:

5.20.1 Flexible copper connections shall be provided between rotating earth blades and the frame which shall have a cross-section of atleast 50 mm. and shall be tinned or suitably treated against corrosion.

5.20.2 The frame of each Isolator and earthing switch shall be provided with two reliable earthing terminals for connection to the Purchaser's earthing conductor/flat so also clamping screws suitable for carrying specified short time current. Flexible ground connections shall be provided for connecting operating handle to the earthing flat. The diameter of clamping screw shall be atleast 12 mm. The connecting point shall be marked with earth symbol.

5.21 Design and Construction:

5.21.1 Full particulars of design, manufacture, jig template and quality control devices developed for manufacture of the equipment offered in respect of but not limited to the following, shall be furnished with drawings and descriptions along with the bid.

i) Contacts, material, current density etc.,

ii) Design of contact pressure.

iii) Contact support and fixing arrangement on isolators.

iv) Turn and twist mechanism, clamps, locks etc.,

v) Bearings, housing of bearings, bushes etc.,

vi) Base plates.

vii) Brass bushes and bearings at various joints.

viii) Operating mechanism etc.

ix) Interlocking devices.

Offers without the above information or with incomplete information shall be treated as non-responsive.

5.21.2 Fasteners:

Nuts, bolts and washers of 5/8" and higher size shall be hot dip galvanised. The bolts used on tapped holes of insulator cap shall be galvanised by centrifuge process to avoid excess deposition of zinc on threads. Nuts, bolts and washers of less than 5/8" size shall be of stainless steel when used on live parts and nickel plated brass in other parts.

5.21.3 Contacts:

Contacts shall be made out hard drawn electrolytic grade copper. Arcing contacts wherever provided shall close first and open last. The contacts surface shall be silver plated and thickness of silver coating not less than 15 microns. Fabrication of contact shall be made with suitable jig to avoid deviations during production. Details of size and shape of contacts, springs, back plate, fixing arrangements design of contact pressure, life of contacts, limit of temperature rise etc., shall be furnished along with the tender. The moving contacts of tubular section shall be so dimensioned as to carry the rated current.

5.21.4 Mounting of Contacts:

The contacts shall rest on a brass block and with initial tension. Suitable device shall be provided **t**o prevent dashing. Fabrication, welding etc., shall be done in suitable jig to avoid deviations during production.

5.21.5 Terminal Pad:

The terminal pad shall be suitable for connection to aluminium terminal connector through bimetallic plate wherever necessary. It shall be made out of Al./electrolytic copper silver plated.

5.21.6 Moving Blade:

Contact surface of moving blade shall be silver plated with the thickness of silver coating not less than 15 microns. The surface shall be wiped during closing and opening operations to remove any film, oxide coating etc., Wiping action shall not cause scouring or abrasion of surfaces.

5.21.7 The rotating insulator shall be mounted on a housing with bearing housing. The housing shall be made of gravity die cast aluminium with smooth surfaces suitably machined for seating the bearings. Two Nos. of bearings with adequate shaft diameter and distance between the bearings shall be provided to avoid wobbling during operations. The bearings shall be of atleast 50 mm. shaft diameter for 220kV, 40mm shaft diameter for 32kV and 25mm shaft diameter for 33kV. The bearings shall be of reputed make eg. SKF, HMT,TATA or equivalent make to be approved by the Purchaser and lubricated for life. All other friction locations shall be provided with bearings, bushes, joints, springs etc., shall be so designed that no lubrication shall be required during service. Complete details of bearings, bushes, housing, greasing etc., shall be furnished with the tender. The earthing switch shaft shall also be provided with necessary bearings.

5.21.8 Tandem pipes shall be provided as per the General requirement of Isolators given in Annexure - III . Base plate of insulators for connection of tandem pipe shall be made out of one piece 10 mm. thick MS plate. Bolt and shackle device shall be used to connect tandem pipe to the base plate.

5.21.9 Down pipes shall be provided as per Annexure – III. The pipe shall be terminated into a suitable swivel or universal type joint between the insulator bottom bearing and the operating mechanism to take care of marginal angular misalignment at site.

5.21.10 Each phase of isolator shall be provided with a rigid base fabricated from steel sections. The base shall be suitable for mounting on support structures. Fabrication, welding etc., shall be done by suitable jig, power press and templates to avoid deviations during production. Details of dimensions of sections, jig templates and device used for production of the base shall be furnished with the tender.

5.22 Terminal Connectors:

5.22.1 The terminal connectors suitable for single Moose (61/3.53mm) ACSR conductor (6 No.s per Isolator) for 33 kV 800 A and 132 kV 800A).for twin Moose (2x61/3.18mm)ACSR conductor (6 Nos per Isolator for 33kV,1200A), single Moose for 220kV 800A Isolator. The connectors should be made of alluminium alloy in conformity with IS:617.

i) Terminal connectors shall be manufactured and tested as per IS:5561.

ii) All castings shall be free from blowholes, surface blisters, cracks and cavities. All sharp edges and corners shall be blurred and rounded off.

iii) No part of a clamp shall be less than 10 mm. thick.

iv) All ferrous parts shall be hot dip galvanised conforming to IS: 2629.

v) For bimetallic connectors, copper alloy liner of minimum thickness of 2 mm. shall be cast integral with aluminium body.

vi) Flexible connectors shall be made from tinned copper/aluminium sheets.

vii) All current carrying parts shall be designed and manufactured to have minimum contact resistance.

viii) Connectors shall be designed to be corona free in accordance with the requirements stipulated in IS:5561.

5.22.2 Wherever necessary bimetallic strip of standard quality shall be used to avoid galvanic corrosion.

5.23 Assembly:

The Isolator shall be fully assembled at the works of the Supplier and operations shall be carried out on it to ascertain that all parts fit correctly and function satisfactorily.

5.24 Painting, Galvanising and Climate Proofing:

5.24.1 All interiors and exteriors of enclosures, cabinets and other metal parts shall be thoroughly cleaned to remove all rust, scales, corrosion, grease and other adhering foreign matter.

5.24.2 All metal surfaces exposed to atmosphere shall be given two primer coats of Zinc chromate and two coats of epoxy paint with epoxy base thinner. All metal parts not accessible for painting shall be made of corrosion resisting material. All machine finished or bright surfaces shall be coated with a suitable preventive compound and suitably wrapped or otherwise protected. All paints shall be carefully selected to withstand tropical heat and extremes of weather with in the limits specified. The paint shall not scale off or wrinkle or be removed by abrasion due to normal handling. Paint inside the metallic housing shall be of anti-condensation type and all external paintings shall be as per shade no. 631 of IS:5

5.24.3 All components shall be given adequate treatment of climate proofing as per IS:3202 so as to withstand corrosive and severe service conditions.

5.24.4 All metal parts not suitable for painting such as structural steel, pipes, rods, levers, linkages, nuts and bolts used in other than current path etc., shall be hot dip galvanised and shall be capable to prevent corrosion in view of the severe climatic conditions.

5.24.5 Complete details of painting, galvanising and climate proofing of the equipment shall be furnished in the bid.

5.24.6 All bolts and nuts are to be hot dip galvanized as per rules at standards and electro plating will not be accepted.

5.24.7 Turn & Twist Mechanism for 220kV Isolators: Turn and twist mechanism shall be provided with adequate locking to avoid opening or loosening by wind, short circuit force etc. on moving blade. The springs shall be made out of stainless steel or phosphor bronze or any other better material to have adequate strength and resilience and shall be encased with grease to avoid exposure to rain. The clamps and plates shall be made out of at least 10 mm thick M.S. plate or flat. Fasteners with nylock

nuts shall be fabricated by tubular gas cutting or milling. The entire mechanism shall be fabricated in suitable jig and template to avoid deviations during production.

5.25 INTERLOCKING

5.25.1 The earth switch shall be mechanically interlocked so that the earth switch can be operated only when the isolator is opened.

5.25.2 In addition to the constructional interlock, isolator and earth switch shall have provision to prevent their electrical and manual operation unless the other interlocking conditions are met. To achieve the interlock condition One No. 220kV DC operated 10A rating interlocking coil shall be provided in motor operated mechanism. The coil shall have 3 N/O + 3 N/C contacts. The interlocking coil shall be suitable for continuous operation from 220 V DC supply and within variation limits stipulated. This interlocking coil should be provided for 420 kV and 245/132 kV & 33KV Isolators and for Earthing Switches of all voltages.

5.25.3 Interlocks are required to be provided for:

i) to prevent opening of the isolators on load;

ii) to prevent closing of the earthing switch when the isolator is closed;

iii) to prevent closing of the line isolator when the earth switch is closed.

The earthing switch shall be operated only when the isolator and the circuit breaker at the HV, MV & LV side(s) of the transformer are opened.

5.25.4 Provision shall also be made for interlocking isolators with circuit breakers so that these are closed before the circuit breaker is closed and opened after the circuit breaker is opened.

The electrical interlocking scheme shall operate on 220 V D.C. supply as specified in the specification. It shall not be possible to operate the isolator either manually or electrically when the 220 V DC core of the electro magnet is not energised.

5.25.6 The details of interlocking system shall be furnished by the Bidder and shall be subject to the purchaser's approval.

6.0 TESTS:

6.1 Type tests :

6.1.1 All the equipment offered, shall be fully type tested by the bidder or his collaborator as per the relevant standards. The date of type test conducted shall be the latest and not earlier than five years. The Bidder shall furnish one set of the type test reports along with the offer.

6.2 Acceptance and Routine Tests :

6.2.1 All acceptance and routine tests as stipulated in the relevant standards shall be carried out by the Supplier in the presence of Purchaser's representative.

6.2.2 Mechanical operation test (routine test) shall be conducted on Isolator (main switch and earth switch) at the Supplier's works.

6.3 Immediately after finalisation of the programme of type/acceptance, routine testing,

the Supplier shall give sufficient advance intimation to the Purchaser to enable him to depute his representative for witnessing the tests.

6.4 Special Tests:

Special tests listed in Annexure II shall be carried out in the presence of Purchaser's representative.

6.5 Test certificates of various items including but not limited to the following, shall be furnished at the time of routine tests:

a) Chemical analysis of copper along with a copy of excise certificate indicating genuine source of procurement of electrolytic grade copper.

b) Bearings

c) Fasteners

d) Universal/swivel joint coupling

e) Gears

f) Auxiliary switch

g) Interlocking devices

h) Terminal block

i) Solid core insulators

j) Any other item.

7.0 INSPECTION:

i) The Purchaser shall have access at all reasonable times to the works and all other places of manufacture, where the Isolators, earth switches and associated equipment are being manufactured and the Supplier shall provide all facilities for unrestricted inspection of the works, raw materials, manufacture of all the accessories and for conducting necessary tests as detailed herein.

ii) The Supplier shall keep the Purchaser informed in advance of the time of starting and of the progress of manufacture of equipment in its various stages so that arrangements could be made for inspection.

iii) The Supplier shall give 15 days (for local Supplies/30 days (for foreign supplies) advance intimation to enable the Purchaser to depute his representative for witnessing acceptance and routine tests. All charges in connection with inspection by Purchasers Personnel such as travel, accommodation and incidentals shall be borne by the bidder.

iv) No material shall be despatched from its point of manufacture unless the material has been satisfactorily inspected , tested and cleared for despatch.

v) The acceptance of any quantity of the equipment shall in no way relieve the Supplier of the responsibility for meeting all the requirements of this specification and shall not prevent subsequent rejection if such equipment are later found to be defective.

8.0 QUALITY ASSURANCE PLAN: Refer Clause-3 of Section-I, General

The Bidder shall invariably furnish following information alongwith his offer, failing which his offer shall be liable for rejection. Information shall be separately given for individual type of the Isolator.

i) Statement giving list of important raw materials, including but not limited to:

a) Copper

b) Steel

c) Springs

d) Bearings

e) Nuts & Bolts

f) Grease

g) Operating mechanism and its components etc.,

ii) Names of sub suppliers for the important raw materials, list of standards according to which the raw materials are tested, list of tests normally carried out on raw materials in presence of Supplier's representative, and copies of test certificates.

iii) Information and copies of test certificates as in (i) and (ii) above in respect of bought out accessories.

iv) List of manufacturing facilities available.

v) Level of automation achieved and list of areas where manual processing still exists.

vi) List of areas in manufacturing process, where stage inspections are normally carried out for quality control and details of such tests and inspections.

vii) List of testing equipments available with Bidder for final testing of equipment and test plant limitation if any, vis-a-vis the type, special, acceptance and routine tests specified in the relevant standards. These limitations shall be very clearly brought out in the specified test requirements.

8.1 The Supplier shall, within 30 days of placement of order, submit the following information to the Purchaser.

i) List of raw materials as well as bought out accessories and the names of sub-suppliers selected from the lists furnished alongwith offer.

ii) Type test certificates of the raw materials and bought out accessories.

iii) Quality Assurance Plan (QAP) with hold points for Purchaser's inspection. The quality assurance plan and hold points shall be discussed between the Purchaser and Supplier before the QAP is finalised.

8.2 The Supplier shall submit the routine test certificates of bought out accessories and central excise passes for raw materials viz. copper and aluminium conductors, lubricating material, gear materials etc., at the time of routine testing of the fully assembled Isolator.

9.0 DOCUMENTATION:

All drawings shall conform to international standards organisation (ISO) 'A' series of drawing sheet/Indian Standards Specification IS:656. All drawings shall be in ink and suitable for micro filming. All dimensions and data shall be in S.I. Units.

9.1 List of Drawings and Documents:

The Bidder shall furnish four sets of following drawings alongwith his offer:

a) General outline and assembly drawings of the Isolator operating mechanism, structure, insulator and terminal connector.

b) Sectional views and descriptive details of items such as moving blades, contacts, arms contact pressure, contact support bearing housing of bearings, balancing of heights, phase coupling pipes, base plate, operating shaft, guides, swivel joint operating mechanism and its components etc.,

c) Loading diagram.

d) Drawings with structure for the purpose of type tests.

e) Schematic drawing.

9.2 The Supplier shall within 1 week of placement of order, submit four sets of final versions of all the above said drawings for Purchaser's approval. The Purchaser shall communicate his comments/approval on the drawings to the Supplier with in two weeks of receipt of drawings. The Supplier shall, if necessary, modify the drawings and resubmit four copies of the modified drawings for Purchaser's approval within two weeks from the date of Purchaser's comments. The Purchaser shall communicate approval with in two weeks of receipt of modified drawings. After receipt of Purchaser's approval, the Supplier shall within three weeks, submit 20 prints and two good quality reproducibles of the approved drawings for Purchaser's use.

9.3 Adequate copies of acceptance and routine test certificates, duly approved by the Purchaser shall accompany the despatched consignment.

9.4 The manufacturing of the equipment shall be strictly in accordance with the approved drawings and no deviation shall be permitted without the written approval of the Purchaser. All manufacturing and fabrication work in connection with the equipment prior to the approval of the drawing shall be at the Supplier's risk.

9.5 Approval of drawings by Purchaser shall not relieve the Supplier of his responsibility and liability for ensuring correctness and correct interpretation of the latest revision of applicable standards, rules and codes of practices.

9.6 Instruction Manuals:

Twenty copies of the erection, operation and maintenance manuals in English be supplied for each type of Isolator one month prior to despatch of the equipment. The manual shall be bound volumes and shall contain all drawings and information required for erection, operation and maintenance of the Isolator including but not

limited to the following particulars:

a) Marked erection prints identifying the component parts of the Isolator as shipped with assembly drawings.

b) Detailed dimensions and description of all auxiliaries.

c) Detailed views of the insulator stacks, metallics, operating mechanism, structure, interlocks, spare parts etc.

10.0 PACKING AND FORWARDING: Refer Clause-4 of Section-I, General

11.0 QUANTITY AND DELIVERY REQUIREMENTS : Refer Clause-5 of Section-I, General

12 SUPERVISION SERVICES : Refer Clause-6 of Section-I, General

13.0 GUARANTEED TECHNICAL PARTICULARS:

The Guaranteed Technical particulars for the equipment being supplied shall be provided with the Bid as specified in this technical specification at Annexure-I**.** The Bids received without guaranteed technical particulars shall be treated as Non-responsive.

Annexure-I
SYSTEM PARTICULARS

Sl.No.	Electrical System data	220 kV Isolator	132kV Isolator	33kV Isolator
1.	System Voltage (kV) rms	220	132	33
2.	Max. Voltage (kV) rms	245	145	36
3.	Lightning impulse withstand voltage (dry & wet) (kVP)	1050	650	170
4.	Power frequency withstand voltage (wet) kVP	460	275	70
5.	Short Circuit Level (kA)	50	40	31.5
6.	Earthing System	Effectively earthed	Effectively earthed	Effectively earthed

Annexure-II
LIST OF SPECIAL TESTS TO BE CARRIED OUT

Sl.No.	Name of the test	Standard to which it conforms
1.	Tests on terminal connectors	IS:5561
2.	Tests on galvanized components	IS:2633

Annexure-III

GENERAL REQUIREMENTS OF 220kV 800A,132kV 800A, 33kV 1200A and 33kV 800A ISOLATOR METALLICS

Sl. No.	Description	Size			
		220kV 800A	132 kV 800A	33kV 2000A	33 kV 800A
1.	Base Channel	150x75	125x65	100 x 50	100 x 50
2.	Elevator Plate/ Elevator Channel	12 mm thick	10 mm thick	100 x 50	100 x 50
3.	Tandem Pipe	2/3 Nos. 32 mm ID Galvanized Class B of Length 10 Mtrs. For MS/ES	2/3 Nos. 32 mm ID Galvanized Class B of Length 7 Mtrs. For MS/ES	1 No. 25mm ID Galvanised class B of length 3.5 mtrs.	1 No. 25mm ID Galvanised class B of length 3.5 mtrs.
4.	Down Pipe	50 mm NB Galva nized Class B	50 mm NB Galva nized Class B	38mm NB galvanized class B	38mm NB galvanized class B
5.	Fixed Contact	32x4 mm 2 Jaws of HDEC	32x4 mm 2 Jaws of HDEC flat	32x5 mm 4 Jaws of HDEC flat	32x4 mm 2 Jaws of HDEC flat silver

		flat silver plated	silver plated	silver plated	plated
6.	Moving arm	HDEC pipe 38mm OD 30mm ID ends silver plated	HDEC pipe 38mm OD 30mm ID ends silver plated	HDEC pipe 50mm OD 34mm ID ends silver plated	HDEC pipe 38mm OD 30mm ID ends silver plated
7.	Earth switch fixed contact	32x4 mm 2 Jaws of HDEC flat silver plated.	32x4 mm 2 Jaws of HDEC flat silver plated.	32x4 mm 2 Jaws of HDEC flat silver plated.	32x4 mm 2 Jaws of HDEC flat silver plated.
8.	Earth switch moving arm	HDEC pipe 38mm OD 30mm ID ends silver plated.	HDEC pipe 38mm OD 30mm ID ends silver plated.	HDEC pipe 38mm OD 30mmID ends silver plated.	HDEC pipe 38mm OD 30mmID ends silver plated.
9.	Jumper Plate	76x12 mm Al. Flat	76x12 mm Al. flat	100 x 20 mm Al. flat	76 x 12 mm Al. flat
10	Type	Turn&Twist	Turn&Twist	Turn&Twist	Turn&Twist

Note: Tolerance of dimension shall be as per IS:9921

Annexure-IV

GENERAL REQUIREMENT OF SOLID CORE INSULATORS

Description	245kV	145 kV	36 kV
No.of insulator units per stack	2	1	1
Height of each basic unit (mm)	1150	1500 ± 2.5	508+ 1
Total height of stack (mm)	2300 ± 3.5	1500 ± 2.5	508 + 1
Pitch circle diameter	4 Nos. holes of 18 mm dia on the base and top suitable for 16 mm tapped for each unit	4 Nos. holes of 18 mm dia on the base and top suitable for 16 mm tapped for each unit	4 Nos. holes tapped 12 mm.
Top (mm)	127	127	76
Bottom (mm)	225	184	76
Total creepage distance minimum (mm)	7595	4495	1116
Mechanical strength required minimum			
Cantilever (kN)	6	6	6
Tension (kN)	110	80	40
Torsion (kN-M)	4.5	4.5	2.5
Compression (kN)	220	160	90

GUARANTEED TECHNICAL PARTICULARS FOR ISOLATOR METALLICS

(Separate for each rating)

(To be filled in by the tenderer)

1) Type/Installation
2) Manufacturer's Name and Country of Manufacture
3) Standards according to which the isolators are manufactured
4) Maximum design voltage at which the isolator can operate (kV)
5) Frequency (Hz)
6) Rated Voltage (kV)
7) Maximum current that can be safely interrupted by the isolator Inductive (A & % PF) Capacitive (A & % PF)
8) Continuous current rating Nominal (Amps) Under site conditions (Amps)
9) Rated short time current
   i) For 3 seconds (kA rms)
   ii) For 1 second (kA rms)
   iii) Rated peak short time current (kVp)
10) Current density at the minimum cross-section of
    a) Moving blade (Amps/Sq.mm)
    b) Terminal pad
    c) Contacts
    d) Terminal Connector
11) Maximum Temp. rise of current carrying parts when carrying rated current continuously (deg.c)
12) Derating factor for specified site conditions

13) Insulation Levels

i) Impulse withstand voltage (kV peak) Phase to Earth Isolating Distance

ii) Switching surge withstand voltage (kV peak) Phase to Earth Isolating Distance

iii) Power Frequency withstand voltage (kV rms) Phase to Earth Isolating Distance

14) Minimum clearance in air:
    i) Between poles (mm)
    ii) Between live parts and earth (mm)
    iii) Between live parts when switch is open:
        a) On the same pole (mm)
        b) Between adjacent poles (mm)

15) Rated mechanical terminal load
    i) Load along the terminal connector side (Kg.)

ii) Load across the terminal connector side (kg.)

16) Torque required to operate the switch in Kg-m.

17) Contact Zone
   i) Horizontal deflection (mm)
   ii) Vertical deflection (mm)
   iii) Total amplitude of longitudinal movement w.r.t. conductor supporting fixed contact (mm)

18) Design and Construction
   i) No. of insulators per pole
   ii) No. of breaks per pole
   iii) Type of closing /Opening mechanism (Horizontal/Vertical break straight etc.)
   iv) Contacts:
      a) Material and grade
      b) Cross-sectional area (mm)
   v) Moving Blades:
      a) Material and grade
      b) Cross-sectional area (mm)
   vi) Contact Support:
      a) Material and size of channel/block
      b) Material and size of plate
   vii) Rain hood - Material grade and size
   viii) Turn and twist mechanism
      a) Material and size of clamps
      b) Material size of springs
      c) Whether springs are encased
   ix) Nuts and Bolts:
      a) Size, material and grade in live parts
      b) Size, material and grade in other parts
   x) Insulator base plate Material and size of plate below insulators
   xi) Bearings:
      a) Material and size of housing
      b) No. of bearings, location and size
   xii) Tandem pipe:
      a) Size class and no. of pipes
      b) Size of shackle, screw
      c) No. of bearings/bush and its material and size
   xiii) Type of interlock
   xiv) Down pipe size and class
   xv) Type of universal/swivel joint
      a) Between bearing and down pipe
      b) Between down pipe and operating mechanism

   xvi) Operating mechanism
      a) Control cabinet
         Material and thickness

Degree of protection
Type size and No. of cable glands
Whether removable gland plate provided

xvii) Base:
a) Size of steel sections used
b) Overall size
c) Total weight

xviii) Terminal Connectors:
a) Clamp Body:
Alloy Composition
Plating if any
Area at air cross -section
b) Bolts and nuts size
Alloy Composition
Tensile strength
c) Type of washers used
d) Materials of braids
e) Temperature rise when carrying rated current at 50 deg.c ambient (deg.c)
f) Weight of each type of clamp (kg).

19.0 List of bought out items

Sl No.	Particulars of Components	Qty	Rating	Make	Type

20.0 Has Bidder included information sought in clause No. 5.27.1 Yes/NO

21.0 List of test certificates furnished

22.0 List of drawings furnished

Name of the firm:
Signature of Bidder.....
Designation & Seal.....
Date.....

OLTC	On Load Tap Change
WTI	Winding Temperature Indicator
OTI	Oil Temperature Indicator
Torr	Torr.
LOI	Letter of Intent
CIF	Cost including Insurance & Freight
CIP	Carriage Insurance Paid
SF6	Sulphur Hexa Flouride
TRV	Transient Recovery Voltage
CTD	Capacitor Tripping Device
ICU	Individual Compressor Unit
NO	Normally Open
NC	Normally Close
sq.mm Al. Cable	Square millimeter Aluminium cable
sq.mm Cu. Cable	Square millimeter Copper cable
MS Flat	Mild Steel Flat
C	Close
O	Open
HRC	High Rupturing Capacity
PVC	Poly Vinyle Chloride
p.f	.Power factor
CT	Current transformer
NCT	Natural Current Transformer
TEFC	Totally enclosed fan cooled
P.U.	Per Unit
ISO	Indian Standard Organization
CB	Circuit Breaker
Auto	Automatic
kg./Sq.Cm	Kilogram per square centimeter
T	Metric Ton
KN	Kilo Newtons